JAMAL MOHAMEDAIN

Mechanical & Mechatronics Engineer | EIT Certified

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LinkedIn Profile · Portfolio
Nationalities: French | Egyptian | Sudanese

Mechanical and Mechatronics Engineer (B.Eng., EIT) with hands-on CAD design experience in CATIA V5 and SolidWorks, plus instrumented test rigs, data analysis and CFD simulation. Designed, drew and built multiple prototypes from scratch, with exposure to CNC machining, 3D printing, assembly and line commissioning. Seeking a mechanical design role in aerospace, defence or industrial equipment, where CATIA V5 design and hands-on prototyping translate directly into product development.

EXPERIENCE

Production & Maintenance Intern
Jul 2024 – Oct 2024

L'Oréal Cairo Plant | Cairo, Egypt

- Developed and implemented an Autonomous Preventative Maintenance (APM) program across seven production lines by conducting detailed mechanical inspection of filler machines, analyzing camshaft assemblies, chain and belt drives, and moving parts, then producing structured inspection and cleaning guides issued automatically via MaintainX CMMS software, reducing unplanned downtime by 48%.
- Participated in the commissioning of a new production line, including LOTO (Lockout/Tagout) procedures and safety protocols.
- Led the renovation of the weighing room: specified a shaking table and carts that removed heavy manual labour from batch preparation and designed a piping system for the room (quick-connect fittings; not implemented, as installation exceeded the internship timeline).

Production & Maintenance Intern
Aug 2020 – Sep 2020

MEGA Food Industries | Sudanese PEPSI Bottler | Khartoum, Sudan

- Gained foundational industrial experience in safety, hygiene standards, machine operation, and maintenance procedures in a high-volume FMCG manufacturing environment.
- Supported production planning and machine maintenance tasks across bottling and packaging lines.

EDUCATION

B.Eng | Mechanical Engineering & Mechatronics
Sep 2022 – Jul 2025

University of Hertfordshire (UH) – Hosted by GAF | Cairo, Egypt

- Graduation project: Wall-Climbing Robot for silo inspection and maintenance.

Foundation Year | Mechanical Engineering
Sep 2019 – Sep 2022

University of Khartoum | Khartoum, Sudan

- Completed first year and was promoted to second year. Studies were repeatedly interrupted by university closures (COVID-19 and the October 2021 coup) before transferring to UH.

IGCSE & AS-Level
Sep 2016 – Jun 2019

Manarat International School | Jeddah, KSA

- Completed IGCSE and AS-Level certifications across a 3-year period.

NOTABLE PROJECTS

Wall-Climbing Robot [Graduation Project]

Individual, self-proposed graduation project: sole designer of a wall-climbing robot for silo inspection, delivered within the university's £75 budget cap. Modelled all parts and the full assembly in CATIA V5 (Part Design, Assembly Design, Generative Shape Design, Drafting), including a propeller built from a NACA 0015 profile. Simulated its airflow in CFD using Ansys Workbench (Fluent CFD), predicting about 5 N of thrust for a 500 g robot. Produced 2D drawings for laser-cut parts and 3D printed complex-shaped ones. Traced an adhesion failure to air leaking through the ground clearance and fixed it with a compliant foam seal. When the printed propellers failed under load, simulated a commercial fan design in CFD and replaced them with a commercial fan blade, adapting the housing to fit. Also handled electronics and control.

Soft Robotic Gripper for DOBOT [Control Systems]

Designed the gripper from scratch in CATIA V5 instead of using the provided design, including its mould. Cast the gripper in silicone using a 3D-printed PLA mould. Controlled it with PID and Adaptive PID algorithms for precision manipulation.

Seaport Crane Gantry System [Control Systems]
Designed in CATIA V5 (Part Design, Assembly Design, Drafting) and built a 3-axis Cartesian gantry rig modelling seaport crane kinematics using aluminum extrusion, encoded DC motors, and an ESP32 microcontroller. Implemented and benchmarked five control algorithms (PID, Adaptive PID, Fuzzy-Like PID, Self-Tuning Fuzzy, MFAC) via MATLAB over Wi-Fi TCP, with Adaptive PID achieving best tracking performance through real-time gain scheduling.

PLC Industrial Palletizing Simulation [Siemens TIA Portal]
Designed an automated industrial palletizing line using a Siemens PLC, programmed with Ladder Diagrams in Siemens TIA Portal and simulated in Factory IO.

HUSKY UGV Mobile Robot [Autonomous Systems]

Team lead. Designed the chassis in CATIA V5 to match the original dimensions, with motor mounts and access to the electronics. Tested forward and inverse kinematics and PID speed control in Gazebo before assembly.

MPPT Solar Tracking System [Energy]

Designed in CATIA V5 and built a solar rig with sun-tracking capability to achieve Maximum Power Point Tracking (MPPT) for optimized energy harvesting for agricultural pumping systems.

Articulated Pick-and-Place Robotic Arm [Embedded Systems]
Designed in CATIA V5 and built, using Arduino UNO and Nano, a wireless multi-DOF pick-and-place arm.

ORC Waste-Heat Recovery [Independent Research]
Researched a compact closed-loop Organic Rankine Cycle to recover exhaust heat from passenger cars and generate electricity.

CERTIFICATIONS

Engineer in Training (EIT) - NCEES FE Mechanical	April 2026
Google — Professional Certificate in Business Intelligence	June 2024
Google — Professional Certificate in Data Analytics	May 2024
UH — Finite Element Analysis Workshop	August 2024
UH — Embedded Systems Design Workshop	July 2023
Jet Solar — Manufacturing & Commissioning of Solar Panels	August 2023
Arab American Vehicle Co. — Car Assembly, QC & Maintenance	June 2023
AOI — CNC Operation	March 2023

SKILLS

Technical	Tools & Software	Languages
Mechanical Design (CAD)	SolidWorks, CATIA V5 (Part Design, Assembly Design, GSD, Drafting, Basic FEA)	Arabic — Native
PLC Programming	Siemens TIA Portal, Factory IO	English — Fluent
Embedded Systems	Arduino, C++, Python	French — Beginner
FEA & CFD	Ansys Workbench	
3D Printing & Prototyping	UltiMaker Cura	
PCB & Circuit Design	Proteus, MATLAB, Simulink	
Commissioning & LOTO	SketchUp	
Data Analytics	Google BI & Analytics, R & RStudio, MS Excel, SQL	